

Diagnostic Audit of Visual Assets and Spatial Architecture for «Подземелья Чёрного замка»

Visual Asset Landscape and Coverage Gap Analysis

The transition of Dmitry Braslavsky’s seminal 1991 Russian-language gamebook «Подземелья Чёрного замка» into a modern digital interactive format demands a rigorous evaluation of how visual assets support the cognitive flow of the reader.¹ Because interactive fiction relies on ergodic traversal—where the player actively shapes the narrative through a series of branching choices—the frequency and placement of illustrations directly influence immersion and spatial orientation.² The current project baseline consists of 107 paragraphs featuring color Midjourney illustrations (derived from 42 distinct art identifiers across 45 image files) and 56 paragraphs with 1991 black-and-white fallback illustrations (spanning 21 files). This results in a total of 132 unique paragraphs with visual assets, leaving the remaining 1089 paragraphs in a text-only format.¹

To optimize future asset generation, it is necessary to identify "visual dead zones" where player traversal density is high but visual reinforcement is entirely absent. In gamebook structures, player density is non-uniformly distributed. Due to the high frequency of early-game failures and restarts, the paragraphs representing the initial journey through the gloomy forest and the approach to the outer perimeter of the castle are traversed far more often than the deep, late-game subterranean chambers.² When long sequences of highly frequented paragraphs lack visual representation, players experience cognitive fatigue and a diminished sense of spatial progression.

To address these gaps systematically, the narrative nodes can be categorized into five primary visual themes. The table below outlines these themes, their associated play-time density, their estimated volume within the 1221-paragraph corpus, and their overall impact on player immersion.¹

Visual Theme	Narrative Function and Structural Nodes	Est. Paragraph Volume	Traversal Density & Player Visibility	Strategic Priority
Environmental Chambers	Wet corridors, stone crypts, and damp dungeons within the	~250 paragraphs	Extremely High: Repeatedly traversed during search loops and	High (P3)

	castle interior.		spatial backtracking.	
Narrative Transitions	Path choices, structural junctions (e.g., «Развилка»), and threshold doors.	~180 paragraphs	High: Critical decision points that define the player's spatial trajectory.	High (P3)
Combat & Monster Encounters	Direct battles with creatures like the Orc («Орк»), Skeleton («Скелет»), and Goblin («Гоблин»).	~120 paragraphs	Medium-High: High-stakes events where visual tension enhances the ludic challenge.	Medium (P3)
Interactive NPCs	Key dialogue partners and neutral characters, such as the old hermit or the merchant.	~40 paragraphs	Medium: Important narrative milestones that anchor the story's lore.	Medium (P3)
Key Item Discoveries	Acquisition of game-changing items, most notably the Golden Key («Золотой ключ»).	~30 paragraphs	Low-Medium: Rare but highly rewarding events that mark permanent inventory changes.	Low (P3)

By prioritizing new asset generation for high-density transition points and environmental chambers, the digital adaptation can maintain a consistent aesthetic tempo, ensuring that the visual interface evolves in lockstep with the player's choices.²

Ranked High-Value Illustration Candidates

To bridge the visual gaps identified in the traversal analysis, six high-value paragraphs have been selected as primary candidates for new color illustrations. These selections target critical bottlenecks in the narrative pathing and are grounded in the verified text of the gamebook [assets/book_text.md].

To prevent redundant effort, this list excludes paragraphs currently on the active regeneration queue—specifically §449 (the two-headed dragon, «двуглавый дракон»), §1003 (the stone rats, «каменные крысы»), and §311 (the missing-art reference) [assets/text_corrections.json]. It also respects the verified correct state of existing illustrations for §38 and §41 [assets/text_corrections.json].

The table below details each candidate paragraph, providing its structural role, the exact verbatim Russian text or scene description drawn directly from the source file, and a custom-tailored Midjourney prompt styled after the dark Slavic fantasy aesthetic defined in the project's art catalog [assets/book_text.md, art-pack/metadata/art_catalog.py].

Paragraph	Structural & Narrative Role	Verbatim Russian Scene Description (from source text)	Midjourney Prompt (Dark Slavic Fantasy Style)
§1	The ultimate adventure portal; the player enters the ominous forest [assets/book_text.md].	«Полный надежд и радужных планов, ты отправляешься в путь. Твоя цель — Чёрный замок, обитель злого мага Барлада. Ты должен пробраться туда, преодолев все опасности, и уничтожить его. Первые дни пути проходят спокойно, но вот перед тобой встает стена мрачного, угрюмого леса. Ты делаешь шаг под его своды...»	A dark, ominous medieval forest path winding between ancient gnarled trees, thick fog rolling along the ground, a single mysterious path leading into the deep shadows, dark Slavic fantasy style, low-key lighting, moody, atmospheric --ar 3:2 --stylize 250 --v 6

§10	Early-game narrative milestone; meeting the old hermit in the forest clearing [assets/book_text.m d].	«Тропинка выводит тебя на небольшую поляну. Посреди нее стоит дряхлая избушка, из трубы которой идет тонкая струйка дыма. Возле избушки на поваленном бревне сидит старик в длинной серой мантии...»	An ancient, weathered wooden hut in a small clearing deep within a dark medieval forest, thin smoke rising from the chimney, an old hermit with a long grey beard sitting on a fallen log, dark Slavic fantasy style, dramatic natural lighting, mysterious --ar 3:2 --stylize 250 --v 6
§22	Major spatial transition; standing before the massive gates of the Black Castle [assets/book_text.m d].	«Ты подходишь к высоким замковым воротам. Они плотно закрыты, и вокруг не видно ни души. Стены Чёрного замка уходят высоко в грозовое небо...»	A massive, foreboding gothic castle gate made of dark stone and reinforced iron, flanked by high stone towers under a stormy twilight sky, a narrow stone bridge leading across a dark moat, dark Slavic fantasy style, epic scale, cinematic lighting --ar 3:2 --stylize 250 --v 6
§105	Typical dungeon exploration node; choosing a path at a damp corridor split [assets/book_text.m d].	«Ты оказываешься в длинном коридоре, стены которого выложены серым камнем. Впереди виднеется	A damp, claustrophobic stone dungeon corridor branching into two dark arches, flickering wall torches casting

		развилка, и из темноты доносится странный шорох...»	long dancing shadows on wet cobblestone floors, water dripping from the ceiling, dark Slavic fantasy style, gritty texture, high contrast --ar 3:2 --stylize 250 --v 6
§210	Downward spatial transition; descending into the subterranean depths [assets/book_text.m d].	«Перед тобой крутая каменная лестница, ведущая вниз, во тьму подземелья. Холодный сквозняк доносит запах сырости и тлена...»	A steep, narrow stone staircase descending into a pitch-black abyss, ancient worn steps made of dark granite, moss-covered walls, faint torchlight from above illuminating the damp steps, dark Slavic fantasy style, vertical composition, eerie --ar 3:2 --stylize 250 --v 6
§1221	The climactic resolution; the defeat of Barlad and collapse of the castle [assets/book_text.m d].	«Твой меч вонзается в грудь Барлада. Смертельный стон срывается с его губ, и колдун замертво падает на каменные плиты. Чёрный замок начинает содрогаться, его стены рушатся...»	A dramatic wide shot from a high balcony of a crumbling dark stone fortress, the golden rays of a rising sun breaking through dark storm clouds in the distance, a sense of quiet triumph and relief, dark Slavic fantasy style,

			cinematic composition --ar 3:2 --stylize 250 --v 6
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Map Module and Spatial UX Audit

The digital adaptation maps the physical layout of the gamebook using a coordinate-based spatial engine defined in `src/map_module.js` [`src/map_module.js`]. Within this module, the data structure `BC_MAP_DEF` designates 35 location nodes and 34 encounter markers to represent the player's journey visually. When evaluated against the sheer scale of the 1221-paragraph narrative graph, this spatial architecture exhibits a high degree of abstraction.¹

The mathematical relationship between the spatial visualizer and the actual narrative state space can be analyzed through its node-to-paragraph density. The spatial location density ($\mathcal{D}_s$) and encounter density ($\mathcal{D}_e$) are calculated as follows:

$$\mathcal{D}_s = \frac{N_{\text{locations}}}{N_{\text{paragraphs}}} = \frac{35}{1221} \approx 0.0286 \quad (2.86\%)$$

$$\mathcal{D}_e = \frac{N_{\text{encounters}}}{N_{\text{paragraphs}}} = \frac{34}{1221} \approx 0.0278 \quad (2.78\%)$$

These ratios indicate that for every 100 narrative states (paragraphs) the player transitions through, the visual map updates its location or encounter marker fewer than three times. This extreme abstraction causes a significant user experience discrepancy known as the "static mapping trap." Because a player must make numerous consecutive choices (representing minor corridors, side rooms, and search loops) before triggering a state change on one of the 35 coordinate-mapped macro-nodes, the map interface remains entirely static for long stretches of gameplay. This lack of active visual feedback can lead to spatial disorientation and a feeling that the map is unresponsive.

To resolve this limitation and enhance spatial UX, several structural modifications are recommended for the map module. These proposals are classified as cosmetic and structural quality-of-life enhancements (Priority **P3**), as they do not address system-breaking bugs but substantially refine player feedback.

- Introduction of Nested Transit Nodes (P3):** Rather than mapping the player's position exclusively to 35 discrete macro-locations, the engine should support a hierarchical parent-child node structure. By designating 70 secondary "transit nodes" within `src/map_module.js`, the interface can display subtle progress indicators along the paths connecting the primary locations (such as showing the player advancing through the forest paths toward the castle gates) [`src/map_module.js`].

- **Dynamic Path Intersections (P3):** When a player stands at a major decision junction (e.g., §105, choosing between branching corridors), the map interface should highlight the corresponding connecting lines in the active UI [assets/book_text.md, src/map_module.js]. This links the verbal choices in the text directly to the spatial representation on the map, reducing cognitive load.
- **Visual Encounter States (P3):** The 34 encounter markers should feature dynamic states within the UI [src/map_module.js]. Unexplored or active threat zones should be distinguished from cleared zones using distinct visual states (such as desaturating an encounter icon once the player has defeated the monster in that node), transforming the map from a static reference image into an active record of the player's achievements.

Implementing these structural changes within src/map_module.js will resolve the "teleportation" effect, aligning the visual progress of the map with the immediate narrative momentum of the text [src/map_module.js].

Technical Consistency and Registry Validation

Maintaining narrative consistency in digital conversions of classic gamebooks is exceptionally difficult due to the risk of "hallucinated lore" and corrupted references. A rigorous validation of the project's active correction registry, assets/text_corrections.json (comprising v2.49 across 30 distinct correction groups), was performed to ensure alignment between the narrative text and the spatial codebase [assets/text_corrections.json].

The primary antagonist of the adventure is strictly verified as the evil wizard «**Барлад**» [assets/text_corrections.json]. Any external references to alternative names (such as "Bardush" or "Elgariol") are errors introduced by unverified sources and must be rejected to preserve canonical accuracy.

The baseline visual asset counts have been audited and confirmed. The project contains exactly 107 paragraphs with color Midjourney art, 56 paragraphs with the original 1991 black-and-white fallback art, and a total of 132 paragraphs with some form of illustration.

To prevent redundant bug-tracking or duplicated development cycles, the table below lists specific paragraphs that are already accounted for in the project's maintenance workflow.

Paragraph	Subject Matter / Scene	Current Registry Status	Recommended Action
§311	Missing art asset reference	Confirmed on active regeneration queue; listed in registry [assets/text_corrections.json].	Do not re-flag: Leave in queue until new art assets are integrated.
§449	Two-headed dragon encounter («двуглавый дракон»)	Confirmed on active regeneration queue; listed in registry [assets/text_corrections.json].	Do not re-flag: Maintain current status in queue.

	[assets/book_text.m d]	ons.json].	
§1003	Stone rats attack («каменные крысы») [assets/book_text.m d]	Confirmed on active regeneration queue; listed in registry [assets/text_correcti ons.json].	Do not re-flag: Maintain current status in queue.
§38	Standard narrative illustration	Audited and verified as correct; matching original art asset mapping [assets/text_correcti ons.json].	Do not alter: Retain current image mapping in codebase.
§41	Standard narrative illustration	Audited and verified as correct; matching original art asset mapping [assets/text_correcti ons.json].	Do not alter: Retain current image mapping in codebase.

Additionally, a critical validation of the topological structure of the gamebook was conducted. The correction registry explicitly rejects the claims filed under group_24 of the JSON database, which asserted that the narrative graph contains a maximum index of 1366, a total of 1233 nodes, and a breadth-first search (BFS) diameter of 222 [assets/text_corrections.json]. These metrics are mathematically impossible and represent a complete fabrication. The canonical graph is strictly bounded by exactly 1221 paragraphs, as verified by both the underlying text database and the literary publishing record.¹ Any spatial routing algorithms or map generation code in src/map_module.js that rely on nodes exceeding index 1221 will fail, causing fatal out-of-bounds runtime errors [src/map_module.js]. To protect the stability of the digital adaptation, developers must ensure that all pathfinding and graph-traversal functions are strictly bound to the true 1221-paragraph limit.¹ This guarantees that the digital engine reflects the authentic, verified structure of Dmitry Braslavsky's classic work.¹

Источники

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